

4. (1) $P(X=k) = (1-p)^{k-1} p = pq^{k-1}, k=1, 2, 3, 4, \dots$

(2) $P(Y=k) = (1-p)^{k-1} p^k = p^k q^{k-1}, k=1, 2, 3, 4, \dots$

(3) 依题意, 则 $p=45\%$ $p = X=2n > = pq^{2n-1} = \left(\frac{9}{20}\right) \times \left(\frac{11}{20}\right)^{2n-1}$
 $= \frac{9}{20}$ $S_n = 0, \frac{1-p^n}{1-p}$

X 取偶数时概率 $P\left\{\bigcup_{n=1}^{\infty} (X=2n)\right\} = \frac{\frac{9}{20} \times \frac{11}{20} \left[1 - \left(\frac{11}{20}\right)^{2n}\right]}{1 - \left(\frac{11}{20}\right)^2} = \frac{0.45 \times 0.55}{1 - 0.55^2} = \frac{11}{31}$

6. (1) 恰有 2 台设备被使用为事件 A

$P(A) = C_5^2 (0.1)^2 (0.9)^3 = 10 \times 0.1 \times \frac{1}{10} \times \frac{9^3}{10^3} = \frac{9^3}{10^4} = 0.0729$

(2) 只有 1 台设备被使用为事件 B $P(B) = C_5^1 (0.1) (0.9)^4 = \frac{5}{10} \times \frac{9^4}{10^4} = \frac{5 \times 9^4}{10^5}$

没有设备被使用为事件 C $P(C) = (0.9)^5 = \frac{9^5}{10^5}$

至少有 3 台设备被使用 D $P(D) = 1 - P(A) - P(B) - P(C)$

$= 1 - \frac{9^3}{10^4} - \frac{1}{2} \times \frac{9^4}{10^4} - \frac{9^5}{10^5} = 0.00856$

(3) 至多有 3 台被使用 E

恰有 3 台使用 F

$P(F) = 1 - P(C_5^3 (0.1)^3 (0.9)^2) = 1 - (0.1)^3 (0.9)^2 = \frac{9^2}{10^4}$

$P(E) = P(A) + P(B) + P(C) + P(F)$

$= \frac{9^3}{10^4} + \frac{5 \times 9^4}{10^5} + \frac{9^5}{10^5} + \frac{9^2}{10^4} = 0.99954$

(4) 至少有一台被使用概率 $P(G) = 1 - P(C) = 1 - (0.9)^5 = \frac{10^5 - 9^5}{10^5} = 0.40951$

记第一次检验就能接受的概率为

9. (1) $P_{(A)} = (0.9)^{10} + C_{10}^1 (0.9)^9 (0.1) + C_{10}^2 (0.9)^8 (0.1)^2 = 0.349$

(2) 需作第二次检验为事件 B $P(B) = 1 - P(A) = (0.9)^{10} + C_{10}^1 (0.9)^9 (0.1) + C_{10}^2 (0.9)^8 (0.1)^2$
 第二次被接受为事件 C $= 0.581$

(3) $P_{(C)} = (0.9)^5 = 0.590$

(4) $P(C|B) = \frac{P(BC)}{P(B)} = \frac{P(C)P(B)}{P(B)} = P(C)$

$P(BC) = P(B)P(C) = 0.343$

(5) $P(A) + P(C) = 0.349 + 0.343 = 0.692$



13. 依题意 $X \sim \pi(\frac{t}{2})$

$$P\{X=k\} = \frac{(\frac{t}{2})^k e^{-\frac{t}{2}}}{k!}$$

(1) 中午12时至下午3时 $t=3$ $k=0$

$$P\{X=0\} = e^{-\frac{3}{2}} = 0.2231$$

(2) 中午12时至下午5时 $t=5$ $k=1$ $P\{X=1\} = \frac{5}{2} e^{-\frac{5}{2}}$

收到一次紧急电话的概率 $1 - P\{X=0\} = 1 - e^{-\frac{5}{2}} = 0.9179$

16. 因为当 n 很大时, p 很小, 以 np 为参数的二项分布的概率值可以由参数 $\lambda=np$ 的泊松分布概率值近似

$\lambda=1$ $X \sim \pi(1)$ $P = \frac{(1)^k e^{-1}}{k!} = \frac{1}{k!} e^{-1}$ $p = \frac{\lambda^k e^{-\lambda}}{k!}$

$P\{X=0\} = \frac{1}{0!} e^{-1}$ $P\{X=2\} = \frac{1}{2!} e^{-1}$

$P\{X=1\} = \frac{1}{1!} e^{-1} \times 1$

出事故的车辆数不小于2 $P(X \geq 2) = 1 - P\{X=0\} - P\{X=1\} = 1 - \frac{2}{e} = \frac{e-2}{e} = 0.009$

19. (1) $P(X \leq 3) = F_X(3) = 1 - e^{-1.2}$ ✓

(2) $P(X > 4) = 1 - P(X \leq 4) = 1 - F_X(4) = 1 - (1 - e^{-1.6}) = e^{-1.6}$

(3) $P(3 < X \leq 4) = 1 - P(X \leq 3) - P(X > 4) = 1 - (1 - e^{-1.2}) - e^{-1.6} = e^{-1.2} - e^{-1.6}$

(4) $P(X \leq 3) + P(X > 4) = 1 - e^{-1.2} + e^{-1.6} = e^{-1.2} - e^{-1.6}$

(5) $P(X=2.5) \Rightarrow f_X(2.5) = \frac{dF_X(x)}{dx} \Big|_{x=2.5} = (0.4 e^{-0.4}) \Big|_{x=2.5}$

连续 $P(X=2.5) = 0$ ✓

分布函数单调不减



24. $f_X(x) = \begin{cases} \frac{1}{5} e^{-\frac{x}{5}}, & x > 0 \\ 0, & \text{其他} \end{cases}$ $F_X(x) = \begin{cases} 1 - e^{-\frac{x}{5}}, & x > 0 \\ 0, & \text{其他} \end{cases}$

单次离开概率 $P(X > 10) = 1 - F_X(10) = 1 - (1 - e^{-\frac{10}{5}}) = e^{-2} = \frac{1}{e^2}$ ✓

$P(Y=k) = \binom{5}{k} (\frac{1}{e^2})^k (1 - \frac{1}{e^2})^{5-k}$ $k=0, 1, 2, 3$

$P(Y=0) = (1 - \frac{1}{e^2})^5$

$P(Y \geq 1) = 1 - P(Y=0) - P(Y=1)$ ≈ 0.516

$P(Y=1) = \frac{5}{e^2} (1 - \frac{1}{e^2})^4$

$= 1 - (1 - \frac{1}{e^2}) (1 - \frac{1}{e^2})^4 - \frac{5}{e^2} (1 - \frac{1}{e^2})^4$
 $= 1 - (1 + \frac{4}{e^2}) (1 - \frac{1}{e^2})^4$



$$X \sim N(3, 2^2) \quad \mu=3 \quad \sigma=2 \quad Z = \frac{X-3}{2} \sim N(0, 1)$$

$$26. (1) P\{2 < X \leq 5\} = P(X \leq 5) - P(X \leq 2) = P(Z \leq 1) - P(Z \leq -\frac{1}{2})$$

$$= P(Z \leq 1) - P(X > \frac{1}{2}) = P(Z \leq 1) - [1 - P(Z \leq \frac{1}{2})]$$

$$= 0.8413 + 0.6915 - 1 = 0.5328$$

$$(2) P\{-4 < X \leq 10\} = P(X \leq 10) - P(X \leq -4) = P(Z \leq \frac{7}{2}) - P(Z \leq -\frac{7}{2})$$

$$X=10 \quad Z = \frac{10-3}{2} = \frac{7}{2}$$

$$X=-4 \quad Z = -\frac{7}{2}$$

$$\text{由于 } Z = \frac{7}{2} > 3 \times 1$$

$$\text{故 } P(10-3=7 > 3 \times 2=3\sigma)$$

根据3σ法则, 其落于此区间几乎肯定

逼近

$$P\{-4 < X \leq 10\} \approx 99.74\% \rightarrow \Phi(3.5) - \Phi(-3.5) = 0.9996$$

$$(3) P\{|X| > 2\} = P(X > 2) + P(X < -2)$$

$$= 1 - P(X \leq 2) + P(X < -2)$$

$$= 1 - P(X \leq 2) + P(X \leq -2) = 1 - P(Z \leq \frac{1}{2}) + P(Z \leq -\frac{5}{2})$$

$$X=2 \quad Z = \frac{2-3}{2} = -\frac{1}{2}$$

$$X=-2 \quad Z = \frac{-2-3}{2} = -\frac{5}{2}$$

$$= 1 - P(Z \leq -\frac{1}{2}) + 1 - P(Z \leq \frac{5}{2})$$

$$= P(Z \leq \frac{1}{2}) + 1 - P(Z \leq \frac{5}{2})$$

$$= 0.6915 + 1 - 0.9938$$

$$= 0.6977$$

$$(4) P\{X > 3\} \quad 3 = \mu$$

$$= 1 - P\{X \leq 3\} = \frac{1}{2}$$

$$(2) P\{X > c\} = P\{X \leq c\} \quad c = \mu = 3$$

$$1 - \Phi(\frac{c-3}{2}) \geq 0.9$$

$$\Phi(-\frac{c-3}{2}) \geq 0.9 =$$

$$\Phi(1.282)$$

$$(3) P\{X > d\} \geq 0.9 \Rightarrow P(X \leq d) \leq 0.1 \quad P(X=2.33) = 0.9901$$

$$1 - P(X \leq d)$$

查表 d 至多取 2.32

$$28. \text{依题意 } X \sim N(10.05, 0.06^2) \quad Z = \frac{X-\mu}{\sigma}$$

$$0.12 = 2 \times 0.06 = 2\sigma$$

$$X = 10.05 + 0.12 \quad Z = \frac{10.05 + 0.12 - 10.05}{0.06} = \frac{0.12}{0.06} = 2$$

$$p = 1 - \Phi(2) + \Phi(-2)$$

$$Z = \frac{10.05 + 0.12 - 10.05}{0.06} = 2$$

$$= 0.0456$$

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = P(-2 \leq Z \leq 2) = P(X \leq 2) - P(X \leq -2)$$

$$= 2[P(X \leq 2) - P(X \leq 0)] = P(X \leq 2) - [1 - P(X \leq 2)] \times 2$$

$$= 2 \times (0.9772 - 0.5)$$

$$= 2 \times 0.4772 = 0.9544$$



30. 依题意得 $X \sim N(120, 2^2)$

$$\begin{aligned} X=122 \quad z &= \frac{122-120}{2} = 1 \\ X=118 \quad z &= \frac{118-120}{2} = -1 \end{aligned}$$

$$P(48 \leq X \leq 22) =$$

$$\begin{aligned} P(-1 \leq X \leq 1) &= 2[P(X \leq 1) - P(X \leq 0)] \\ &= 2 \times (0.8413 - 0.5) \\ &= 2 \times 0.3413 \\ &= 0.6826 \end{aligned}$$

$$P(X \leq -1) + P(X \geq 1) = 1 - P(-1 \leq X \leq 1) = 0.3174$$

有2次测定值落在区间 $[118, 122]$ 之间的概率为 $P(A)$, k 次落在区间内

$$\begin{aligned} A \sim B(3, 0.3174) \quad P(A) &= P(k=2) = C_3^2 (0.3174)^2 (0.6826)^3 \\ &= 0.3204 \end{aligned}$$

X 在 $(0, 1)$ 内服从均匀分布, 则 $f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{其他} \end{cases}$ 分别记 X, Y 的分布函数为 $F_X(x), F_Y(y)$

$$\begin{aligned} 34. (1) Y = e^X \quad F_Y(y) &= F_X(e^X) P\{Y \leq y\} = P\{e^X \leq y\} \\ &= P(X \leq \ln y) = F_X(\ln y) \end{aligned}$$

$$F_Y(y) = F_X(\ln y) \quad f_Y(y) = f_X(\ln y) (\ln y)'$$

$$\begin{aligned} f_Y(y) &= \begin{cases} \frac{1}{y}, & 0 < \ln y < 1 \\ 0, & \text{其他} \end{cases} \\ &= \begin{cases} \frac{1}{y}, & 0 < \ln y < 1 \\ 0, & \text{其他} \end{cases} \end{aligned}$$

$$\begin{aligned} (2) Y = -2 \ln X \quad F_Y(y) &= P\{Y \leq y\} = P\{-2 \ln X \leq y\} \\ &= P\{\ln X \geq -\frac{y}{2}\} = P\{X \geq e^{-\frac{y}{2}}\} \\ &= 1 - P(X < e^{-\frac{y}{2}}) \\ &= 1 - F_X(e^{-\frac{y}{2}}) \end{aligned}$$

$$\begin{aligned} f_Y(y) &= -f_X(e^{-\frac{y}{2}}) (e^{-\frac{y}{2}})' \\ &= -f_X(e^{-\frac{y}{2}}) (e^{-\frac{y}{2}}) (-\frac{1}{2}) \\ &= \frac{1}{2} f_X(e^{-\frac{y}{2}}) e^{-\frac{y}{2}} \\ &= \begin{cases} \frac{1}{2} e^{-\frac{y}{2}}, & 0 < e^{-\frac{y}{2}} < 1 \\ 0, & \text{其他} \end{cases} \end{aligned}$$

$$\begin{aligned} -\frac{y}{2} \leq 0 \quad y \geq 0 \\ f_Y(y) &= \begin{cases} \frac{1}{2} e^{-\frac{y}{2}}, & y \geq 0 \\ 0, & \text{其他} \end{cases} \end{aligned}$$



35. (1). $y < 0$, 则 $f_Y(y) = 0$, $y > 0$, $X \sim N(0, 1)$

故 Y 的分布函数为

$$\begin{aligned} F_Y(y) &= P\{Y \leq y\} = P\{0 < Y \leq y\} = P\{0 < e^X \leq y\} \\ &= P\{-\infty < X \leq \ln y\} \\ &= P\{X \leq \ln y\} = \Phi(\ln y) \end{aligned}$$

$$y > 0, f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dx} \Phi(x) \Big|_{x=\ln y} \cdot \frac{1}{y} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(\ln y)^2} \cdot \frac{1}{y}$$

$$f_Y(y) = \begin{cases} 0, & y \leq 0 \\ \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(\ln y)^2} \frac{1}{y}, & y > 0 \end{cases}$$

(2) $Y = 2X^2 + 1 \geq 1$

$$\begin{aligned} F_Y(y) &= P\{Y \leq y\} = P\{0 < Y \leq y\} \\ &= P\{Y \leq y\} - P\{Y \leq 1\} \\ &= P\{2X^2 + 1 \leq y\} - P\{2X^2 + 1 \leq 1\} \end{aligned}$$

$P\{X=0\} = 0$

$$= P\left\{\sqrt{\frac{y-1}{2}} \leq X \leq \sqrt{\frac{y-1}{2}}\right\} - P\{X=0\}$$

$$= \Phi\left(\sqrt{\frac{y-1}{2}}\right) - \Phi\left(-\sqrt{\frac{y-1}{2}}\right) = P\{X \leq \sqrt{\frac{y-1}{2}}\} - P\{X \leq -\sqrt{\frac{y-1}{2}}\} = 2P\{X \leq \sqrt{\frac{y-1}{2}}\} - 1$$

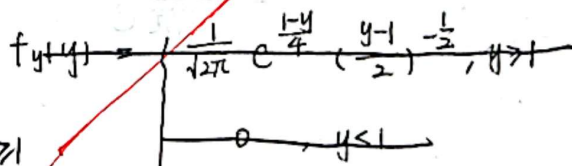
$$f_Y(y) = \left[\Phi'\left(\sqrt{\frac{y-1}{2}}\right) \frac{1}{2} \left(\frac{y-1}{2}\right)^{-\frac{1}{2}} \cdot \frac{1}{2}\right] + \left[\Phi'\left(-\sqrt{\frac{y-1}{2}}\right) \frac{1}{2}\right]$$

$$= \frac{1}{2} \Phi'\left(\sqrt{\frac{y-1}{2}}\right) \left(\frac{y-1}{2}\right)^{-\frac{1}{2}} \cdot \frac{1}{2} + \frac{1}{2} \Phi'\left(-\sqrt{\frac{y-1}{2}}\right) \frac{1}{2}$$

$$y \geq 1, f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-1}{2}\right)} \left(\frac{y-1}{2}\right)^{-\frac{1}{2}} \cdot \frac{1}{2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-1}{2}\right)} \left(\frac{y-1}{2}\right)^{-\frac{1}{2}} \cdot \frac{1}{2} = \frac{1}{\sqrt{2\pi}} \frac{e^{-(y-1)/4}}{2\sqrt{\pi(y-1)}}$$

$y < 1$ $y < 1$ $f_Y(y) = 0$

$$f_Y(y) = \begin{cases} \frac{e^{-(y-1)/4}}{2\sqrt{\pi(y-1)}}, & y \geq 1 \\ 0, & y < 1 \end{cases}$$



(3) $Y = |X| \geq 0$ $y < 0$ $f_Y(y) = 0$

$$F_Y(y) = P\{Y \leq y\} = P\{|X| \leq y\} = P\{-y \leq X \leq y\} = 2P\{X \leq y\} - 1$$

$$f_Y(y) = 2f_X(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \times 2 = \frac{2}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} = \frac{2}{\sqrt{\pi}} e^{-\frac{y^2}{2}}$$

$$f_Y(y) = \begin{cases} \frac{2}{\sqrt{\pi}} e^{-\frac{y^2}{2}}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$



$$36. (1) Y = X^3 \quad F_Y(y) = P(X \leq y) = P(X^3 \leq y) = P(X \leq y^{\frac{1}{3}}) \\ = F_X(y^{\frac{1}{3}}) \\ f_Y(y) = f_X(y^{\frac{1}{3}}) \cdot \frac{1}{3} y^{-\frac{2}{3}}$$

$$(2) f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{其他} \end{cases}$$

$$Y = X^2 \quad F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ = P(X \leq \sqrt{y}) - P(X \leq -\sqrt{y})$$

$$f_Y(y) = f_X(\sqrt{y}) \cdot \frac{1}{2} y^{-\frac{1}{2}} - f_X(-\sqrt{y}) \cdot (-\frac{1}{2}) y^{-\frac{1}{2}} = \frac{1}{2} f_X(\sqrt{y}) y^{-\frac{1}{2}} + \frac{1}{2} f_X(-\sqrt{y}) y^{-\frac{1}{2}} \\ = \frac{1}{2} [e^{-\sqrt{y}} + e^{-(-\sqrt{y})}] y^{-\frac{1}{2}} = \frac{1}{2} f_X(\sqrt{y}) y^{-\frac{1}{2}} + \frac{1}{2} f_X(-\sqrt{y}) y^{-\frac{1}{2}}$$

$$39. \theta = \frac{5}{9}(T-22), T \sim N(98, 6, 2)$$

由例4. 随机变量 $X \sim N(\mu, \sigma^2)$, X 的线性函数 $Y = aX + b$ 也服从正态分布.

$$f_T(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(t-98.6)^2}{4}}, -\infty < t < \infty \\ \theta \sim N\left(\frac{5}{9}(98-22), \left(\frac{5}{9} \times 6.2\right)^2\right) \\ f_\theta(\theta) = \frac{9}{5} f_T\left(\frac{\theta - \frac{5}{9} \times 22}{\frac{5}{9}}\right) \\ = \frac{9}{5} f_T\left(\frac{9}{5}\theta - 22\right)$$

将 θ 的分布函数记为 $F_\theta(y)$, 则有

$$F_\theta(y) = P\{\theta \leq y\} = P\left\{\frac{5}{9}(T-22) \leq y\right\} = P\{T \leq \frac{9}{5}y + 22\} = \int_{-\infty}^{\frac{9}{5}y+22} f_T(t) dt$$

$$f_\theta(y) = \frac{9}{5} f_X\left(\frac{9}{5}y + 32\right) = \frac{1}{\sqrt{2\pi}} \frac{9}{5} e^{-\frac{(\frac{9}{5}y - 66.6)^2}{4}} \\ = \frac{9}{10\sqrt{\pi}} e^{-\frac{81}{100}(y-3)^2}$$

